



Django Models & Relationships Quick Notes

Video Reference: Chai aur Code - Django Relationship Models (Chai Project) • **Language:** Hinglish Super Easy Guide

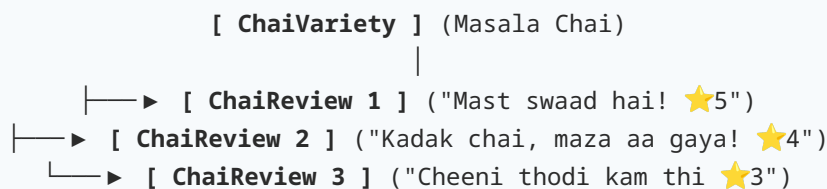
🌟 Introduction: Yeh Notes Kis Baare Mein Hai?

Is video mein Hitesh sir ne **Django Models ke 3 main Relationships** aur unko **Django Admin Panel** ke andar mast visual style mein configure karna sikhaya hai:

Relationship	Real Life Chai Example	Django Field Type
One-to-Many (1:N)	1 Chai ke multiple Reviews ho sakte hain (par ek review ek hi specific chai ka hoga).	<code>models.ForeignKey</code>
Many-to-Many (M:N)	1 Chai multiple Stores pe mil sakti hai, aur 1 Store mein multiple Chai varieties bik sakti hain.	<code>models.ManyToManyField</code>
One-to-One (1:1)	1 Chai ka ek hi Unique FSSAI Certificate hoga, aur 1 Certificate ek hi Chai ko belong karega.	<code>models.OneToOneField</code>

1 One-to-Many (1 : N) Relationship - ForeignKey

Simple Bhasha mein: Jab "Parent" ek ho aur uske "Children" bohot saare ho sakein!



Code Implementation (models.py):

```
from django.db import models
from django.contrib.auth.models import User
from django.utils import timezone

class ChaiReview(models.Model):
    # Foreign key batata hai ki ye review kis chai ka hai
    chai = models.ForeignKey(ChaiVariety, on_delete=models.CASCADE, related_name='reviews')
    user = models.ForeignKey(User, on_delete=models.CASCADE)
    rating = models.IntegerField()
    comment = models.TextField()
    date_added = models.DateTimeField(default=timezone.now)

    def __str__(self):
        return f'{self.user.username} review for {self.chai.name}'
```

🔑 Important Keywords:

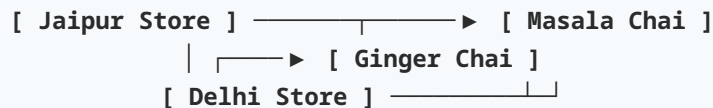
- `on_delete=models.CASCADE`: Agar main chai delete ho gayi, toh uske saare reviews bhi database se apne-aap

delete ho jayenge!

- `related_name='reviews'`: Ab hum chai object se direct reviews access kar sakte hain:
`chai_obj.reviews.all()`.

2 Many-to-Many (M : N) Relationship - `ManyToManyField`

Simple Bhasha mein: Dono taraf se bohot saare connections!



(Jaipur store par Masala & Ginger dono hain, aur Masala Chai Jaipur & Delhi dono store pe hai!)

Code Implementation (`models.py`):

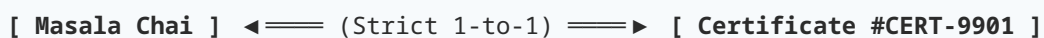
```
class Store(models.Model):
    name = models.CharField(max_length=100)
    location = models.CharField(max_length=100)
    # Many to many connection with ChaiVariety
    chai_varieties = models.ManyToManyField(ChaiVariety, related_name='stores')

    def __str__(self):
        return self.name
```

 **Under the Hood:** Django peeche se ek choti teesri table (Bridge/Junction Table) create karta hai jo dono tables ki IDs ko map karti hai.

3 One-to-One (1 : 1) Relationship - `OneToOneField`

Simple Bhasha mein: Ek item ke liye sirf aur sirf ek hi item allow hoga!



Code Implementation (`models.py`):

```
class ChaiCertificate(models.Model):
    chai = models.OneToOneField(ChaiVariety, on_delete=models.CASCADE,
    related_name='certificate')
    certificate_number = models.CharField(max_length=100)
    issued_date = models.DateTimeField(default=timezone.now)
    valid_until = models.DateTimeField()

    def __str__(self):
        return f'Certificate for {self.chai.name}'
```

Admin Panel Superpowers & Customization (admin.py)

Hitesh sir ne Django Admin ko powerful banane ke liye 3 zabardast tools sikhaye:

1. TabularInline (Chai ke form ke andar hi Reviews show karna):

Alag se review page kholne ki zaroorat nahi, Chai open karte hi neeche reviews add karne ke 2 dabbe dikhenge!

```
class ChaiReviewInline(admin.TabularInline):
    model = ChaiReview
    extra = 2 # By default 2 blank review rows dikhengi
```

2. ModelAdmin & list_display (Columns customize karna):

```
class ChaiVarietyAdmin(admin.ModelAdmin):
    list_display = ('name', 'type', 'date_added') # Table columns
    inlines = [ChaiReviewInline] # Inline reviews attach kiya
```

3. filter_horizontal (Many-to-Many Multi-Select Box):

Store admin panel mein left side saari chai varieties aur right side selected chai ka stylish UI box banta hai.

```
class StoreAdmin(admin.ModelAdmin):
    list_display = ('name', 'location')
    filter_horizontal = ('chai_varieties',) # Dual multi-select widget
```

4. Registering in Admin:

```
admin.site.register(ChaiVariety, ChaiVarietyAdmin)
admin.site.register(Store, StoreAdmin)
admin.site.register(ChaiCertificate, ChaiCertificateAdmin)
```

Golden Rules & Terminal Commands to Remember

- Jab bhi models.py mein naya model ya field banao:**
python manage.py makemigrations (*Blueprint banta hai*)
python manage.py migrate (*Database table banti hai*)
- Admin Tuple Trailing Comma:**
Agar tuple mein ek hi item hai, toh comma lagana zaroori hai: filter_horizontal = ('chai_varieties',)
nahi toh Python error dega!
- Folder Path Rule:** Terminal hamesha project folder (jahan manage.py hai) mein hona chahiye, virtual environment ke andar nahi!